Panskura Banamali College (Autonomous)

B.Sc.(H) 4th Semester Examination - 2022

Sub: BCA **Paper: CC10T**

Paper Name: Data Communication & Computer Network

F.M.: 40 **Time : 2 Hours**

Answer any four questions: **4x10=40**

1. a) Explain the functions performed by different layer of ISO-OSI model.
   b) Compare ISO/OSI model with TCP/IP model. (7+3)
2. a) Differentiate between circuit switched network and packet switched network.
   b) What do you mean by Cyclic Redundancy Check (CRC)?
   c) What is piggybacking? (5+3+2)
3. a) Describe different types of classes in IPV4.
   b) Differentiate between classful and classless addressing.
   c) Given IP address: 201.14.78.65 and the subnet mask: 255.255.255.224. What is the subnet address?
   d) What do you mean by logical address? (3+3+2+2)
4. a) Construct the Hamming code for the bit sequence 10011001.
   b) Briefly describe the frame format of a HDLC frame. (5+5)
5. a) What do you mean by adaptive routing?
   b) Describe any one of the adaptive routing algorithms.
   c) How does CSMA/CD handle a collision? (2+4+4)
6. a) What is synchronous transmission?
   b) How does asynchronous transmission differ with synchronous transmission?
   c) Compute the bit rate of a 1000 baud 16-QAM signal.
   d) What is Amplitude Modulation (AM)? (2+3+3+2)